



Development and validation of a parsimonious AI-based mortality risk score for heart failure



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Risk stratification in heart failure (HF) supports clinical decisions, yet existing tools face adoption barriers: conventional scores show modest discrimination and depend on specialised tests (i.e., echocardiography), while artificial intelligence (AI) models require rich longitudinal data and infrastructure. Both approaches face challenges in workflow integration, interpretability, and demonstrating clinical utility. Using 373,389 adults with HF from the UK Clinical Practice Research Datalink (CPRD) Aurum, we developed and validated an AI-based risk model for all-cause mortality and compared it against the MAGGIC score adapted for EHR data (MAGGIC-EHR). Predictive signals from a Transformer model trained on electronic health records (EHRs) were distilled, via SHAP-based feature selection, into a parsimonious 11-variable model, SIMPLE-HF (Simplified Intelligent Mortality Prediction for Longitudinal EHRs in HF patients), using point-of-care variables including age, body mass index, comorbidities, and medications. For 12-month all-cause mortality, SIMPLE-HF achieved superior discrimination (C-index 0.801, 95% CI 0.795–0.806) versus MAGGIC-EHR (0.735, 0.728–0.741) and was well calibrated. At a 0.60 high-risk threshold, SIMPLE-HF identified 177 true events per 1000 screened versus 89 for MAGGIC-EHR. Performance gains were consistent for secondary outcomes. This approach may help bridge advanced AI and routine practice, though external validation and prospective evaluation are needed before clinical adoption.

Heart failure (HF) represents a significant global public health challenge, with high rates of hospitalisation and mortality¹. Risk stratification can support HF management by helping to guide therapeutic decisions, allocate resources, and inform prognosis, alongside, rather than replacing, clinical judgement. This is especially important as the projected population with HF in the UK is expected to double by 2040². While numerous prognostic tools exist, including the Meta-Analysis Global Group in Chronic Heart Failure (MAGGIC) score³, they often present modest discriminatory power and necessitate investigations such as echocardiography for estimation of left ventricular ejection fraction (LVEF). LVEF, while integral to the initial HF diagnostic work-up, is not consistently recorded in routine primary care and has to be re-measured when patients are seen at subsequent timepoints. This limits applicability of these scores when assessing patients at later stages of their disease trajectory as recommended by guidelines⁴. Such limitations likely contribute to their underutilisation in routine clinical practice, despite

guideline recommendations underscoring the need for accurate, yet practical and parsimonious, prognostic models^{5,6}.

Over the past decade, access to extensive healthcare datasets, including electronic health records (EHRs), has expanded considerably^{7,8}. These rich data sources offer comprehensive, longitudinal capture of patient health trajectories, providing unprecedented opportunities to unravel the complex interplay of factors influencing chronic diseases like HF^{9–12}. Concurrently, sophisticated artificial intelligence (AI) methodologies, notably Transformer-based architectures, have demonstrated superior discrimination and calibration in predicting clinical outcomes by modelling complex, latent associations within extensive longitudinal EHR data^{13,14}.

However, the advanced capabilities of these state-of-the-art AI models present their own set of challenges. Their reliance on rich, longitudinal datasets and substantial computational resources often restricts their applicability in settings with limited data availability or infrastructure^{15–17}. This gap highlights a critical unmet need: to harness the predictive power

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demonstrated by complex AI systems without inheriting their barriers to implementation

To this end, we aimed to balance predictive accuracy with ease of use, thereby offering a novel pathway to improve risk stratification in diverse clinical settings. We derived and validated an AI-based parsimonious risk score model. The generalisability of this approach was also explored through validation on secondary outcomes, including rehospitalisation and subsequent major cardiovascular events.

Results
Overview

We developed and validated an artificial intelligence (AI)-based risk model for all-cause mortality in patients with heart failure (HF), using routinely collected primary care data from the UK Clinical Practice Research Datalink (CPRD) Aurum linked to hospital and mortality records⁸. From an open cohort of 373,389 adults with incident HF across 1442 English general practices, models were derived and validated using a practice-level 80/20 split (77,712 patients in the validation set). Within a neural network-based survival framework, we began from the MAGGIC predictor set, augmented it with comorbidity signals surfaced by a complex Transformer model (TRisk)^{13,14}, and applied SHAP-based feature selection to distill these into a parsimonious 11-variable model, SIMPLE-HF (Simplified Intelligent Mortality Prediction for Longitudinal EHRs in HF patients), that relies only on variables available at the point of care. Performance was benchmarked against an EHR-adapted MAGGIC model³ (MAGGIC-EHR) and assessed through discrimination, calibration, net benefit, and clinical-impact analysis for 12- and 36-month all-cause mortality, with secondary analyses of cardiovascular events and hospitalisation.

Population characteristics

Among 19,094,480 individuals contributing to data and aged over 18 years within the 01/01/2010 and 31/12/2020 study period, we identified 373,389 patients with incident HF (Fig. S1) for analysis (77,712 patients in validation) with median follow-up of 7 months (interquartile interval [IQI]: [2, 20]). The median age was 80 years (IQI: [70,87]) and approximately, one-third of the cohort had a history of diabetes, stroke, or myocardial infarction along with more than half of the cohort being prescribed beta-blockers. Approximately, 42% of patients died in five years following baseline (Table 1).

Analyses

We compared SIMPLE-HF against an EHR-adapted MAGGIC benchmark for 12-month all-cause mortality in the held-out validation cohort, assessing discrimination (C-index, AUPRC), calibration (ICI), and net benefit.

For 12-month all-cause mortality, SIMPLE-HF achieved higher C-index (0.801; 95% CI [0.795,0.806]) than MAGGIC-EHR (0.735; [0.728,0.741]) along with higher AUPRC (0.684 [0.677,0.691] vs 0.548 [0.541, 0.555]) (Fig. 1, S4, Table S3). Across most subgroups, SIMPLE-HF’s discrimination remained closer to its overall cohort C-index than MAGGIC-EHR; small differences were captured in some subgroups (e.g., <60 years of age, patients with Atrial Fibrillation subgroups) (Fig. 1). Both models were well-calibrated as shown with their closeness of their calibration curves to the reference curve and low ICI estimates (Fig. 2A, Table S4). Decision curve analysis showed that SIMPLE-HF provided greater net benefit than other strategies across the spectrum of clinically relevant thresholds (up to ~0.6) (Fig. 2B). Analyses for 36-month predictions showed similar patterns of discrimination and calibration estimates (Fig. S5, Table S5).

In impact analysis, SIMPLE-HF increases true positives by 39% while reducing false negatives by 36% for 12-month prediction as compared to MAGGIC-EHR (Fig. 3; Table S6). Higher true positives and fewer false negatives directly translated into SIMPLE-HF outperforming MAGGIC-EHR in both PPV and sensitivity, showing increases of 0.072 and 0.188, respectively, for 12-month prediction. Similar results were found for 36-month prediction (Fig. S6; Table S6).

Table 1 | Population characteristics for derivation and validation datasets in UK data study at baseline

	Derivation (295,677 patients)	Validation (77,712 patients)
Women (%)	142,345 (48.1)	37,427 (48.2)
Median age (years) (IQI)	80 (70, 87)	80 (70, 87)
Smoking status ^a		
Current-Smoker (%)	32,579 (11.0)	9002 (11.6)
Not Current-Smoker (%)	263,098 (89.0)	68,710 (88.4)
Median BMI (kg/m ²) (IQI) ^a	27.4 (24.3, 31.2)	27.5 (24.4, 31.4)
Median SBP (mmHg) (IQI) ^a	139.3 (130.3, 147.3)	139.2 (130.3, 147.3)
Median sodium (mmol/L) (IQI) ^a	139.8 (138.0, 141.1)	139.7 (138.0, 141.1)
Median creatinine (μmol/L) (IQI) ^a	88.9 (75.6, 106.3)	88.6 (75.5, 106.1)
<18 months since incident HF (%)	208,977 (68.7)	68,279 (68.7)
NYHA Classification ^a		
I (%)	78,229 (26.5)	20,590 (26.5)
II (%)	136,531 (46.2)	35,764 (46.0)
III (%)	73,986 (25.0)	19,552 (25.2)
IV (%)	6931 (2.3)	1806 (2.3)
Disease history		
Diabetes (%)	82,385 (27.9)	22,234 (28.6)
COPD (%)	76,553 (25.9)	20,622 (26.5)
AF (%)	146,186 (49.4)	38,039 (48.9)
Stroke (%)	68,756 (23.3)	18,279 (23.5)
MI (%)	81,115 (27.4)	21,414 (27.6)
Prescription history		
Beta-blockers (%)	195,563 (66.1)	52,434 (67.5)
ACE-Is/ARBs (%)	182,411 (61.7)	49,399 (63.6)
Procedure history		
PCI (%)	31,221 (10.6)	8244 (10.6)
CABG (%)	14,940 (5.1)	3869 (5.0)

% percent, IQI interquartile interval (25th and 75th percentiles), BMI body mass index, SBP systolic blood pressure, NYHA New York Heart Association, COPD Chronic obstructive pulmonary disease, AF atrial fibrillation, MI myocardial infarction, ACE-I Angiotensin-converting-enzyme inhibitors, ARB angiotensin receptor blockers, PCI Percutaneous coronary intervention, CABG coronary artery bypass graft.

^aIndicates missing variables; BMI (9.6% missingness), smoking status (4.5%), SBP (4.3%), creatinine (7.1%), NYHA (95.6%), sodium (7.4%). Missing Values imputed through MICE (Multiple Imputation by Chained Equations).

To gauge real-world impact, we examined true-positive and false-negative capture at the 50% threshold and across clinically relevant thresholds per 1000 patients screened.

The clinical-impact curves (Fig. 4) illustrate the number of patients, per 1000 screened, who would be classified as high-risk and who would subsequently experience the outcome across a range of decision thresholds. SIMPLE-HF consistently outperforms MAGGIC-EHR. The contrast is most striking at higher clinically relevant thresholds, such as 0.60, where SIMPLE-HF identifies 177 true events among 257 flagged patients, while MAGGIC-EHR captures only 89 events among 145 flagged patients—half

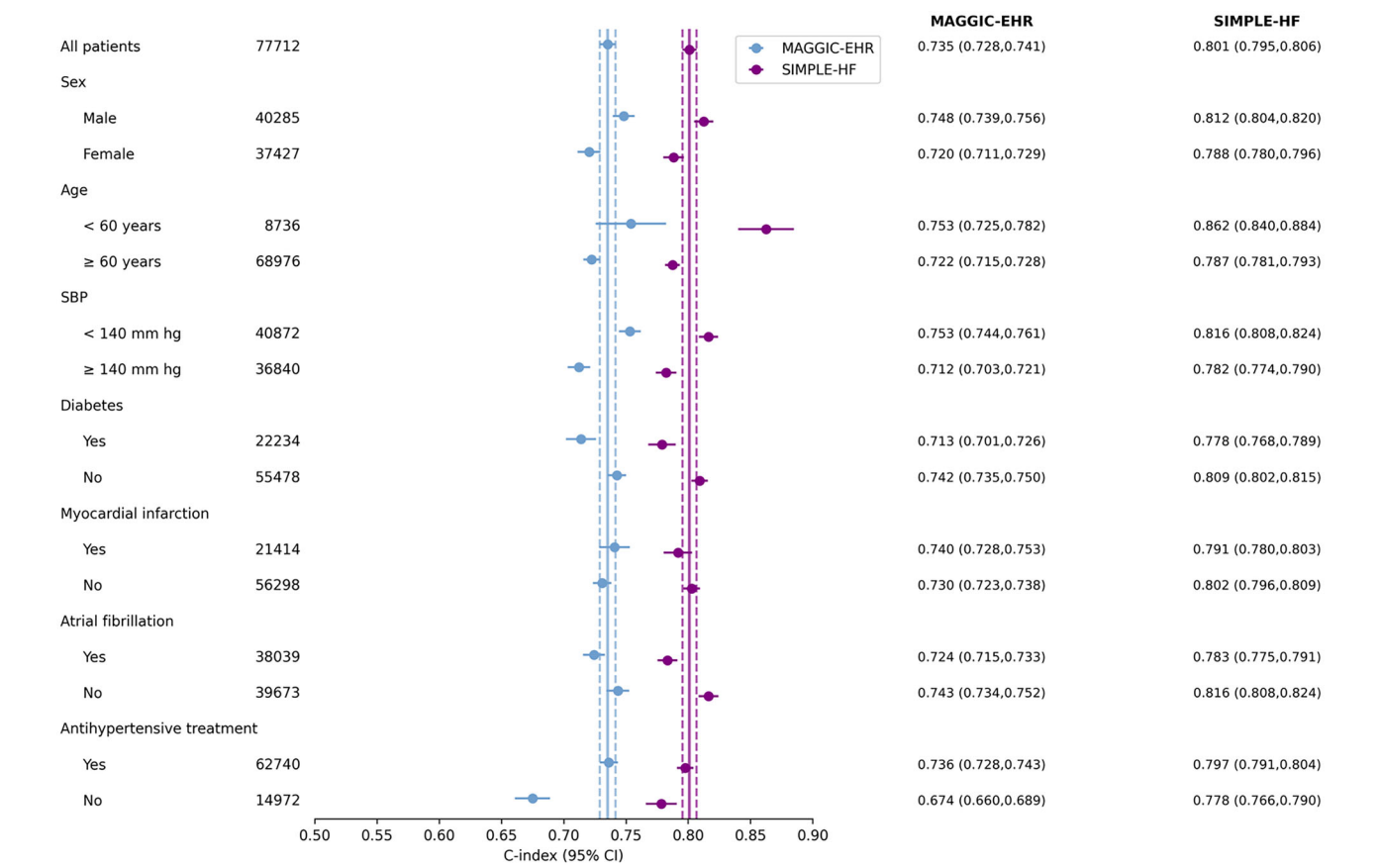


Fig. 1 | Models’ discrimination by concordance index (C-index) and associated 95% confidence intervals (CI) in overall cohort and subgroups for 12-month all-cause mortality risk prediction on UK validation data. Blue and Purple lines represent C-index on “all patients” in the validation cohort for MAGGIC-EHR and SIMPLE-HF, respectively. These lines are provided to visually demonstrate deviation in subgroup discrimination performance from overall cohort performance for each model.

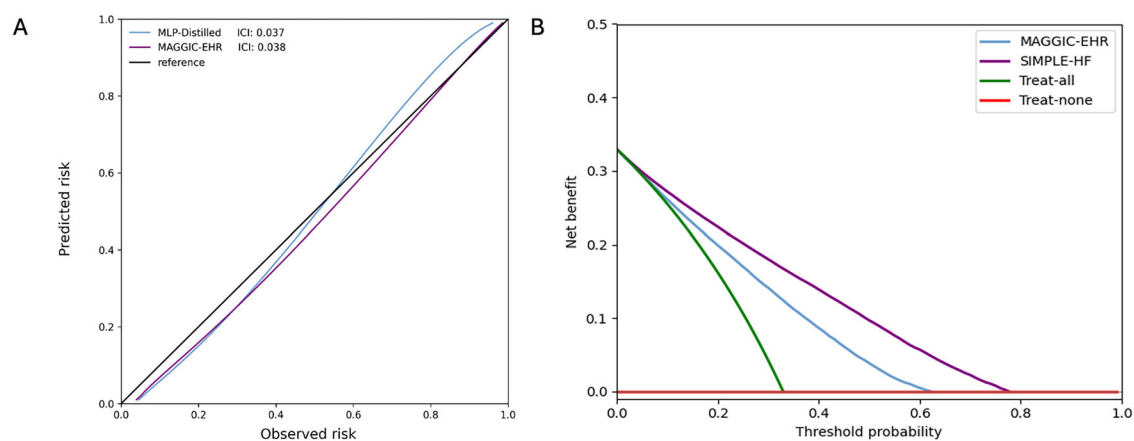


Fig. 2 | Calibration curves and net benefit analysis for 12-month risk prediction of all-cause mortality in UK validation data. Calibration curves with integrated calibration indices (ICI) (A) and decision curve analysis (B) are presented for all-cause mortality at 12-month timepoint. For (A) ICI, lower is better with reference (black line) presenting optimal ICI of 0.0.

the true event capture of SIMPLE-HF. This pattern holds at every threshold examined, demonstrating superior clinical yield with fewer unnecessary alerts. The results persisted for 36-month prediction as well (Fig. S7).

Analyses of secondary outcomes

In addition to our primary outcome of all-cause mortality, we conducted analyses on secondary outcomes for the prediction fatal and non-fatal

cardiovascular outcomes (median follow-up of 4 months — IQI: [1, 14]) and hospitalisation (median follow-up of 4 months IQI: [1, 12]). This was done by applying the same 11-variable feature set, without re-running feature discovery, to predict cardiovascular events and hospitalisation.

These secondary outcome analyses showed similar patterns to the primary outcome, with SIMPLE-HF outperforming MAGGIC-EHR for 12- and 36-month prediction. (Figs. S8-S12, Table S5).

Fig. 3 | Impact analyses at the 50% decision threshold for 12-month all-cause mortality prediction on UK validation data. Sankey diagrams compare predicted outcomes between the models, showing how patient classification compares to actual outcomes, “Events” (i.e., death) and “No events” categories at 50% threshold (denoted as dark blue and red, respectively). TP true positive, TN true negative, FP false positive, FN false negatives.

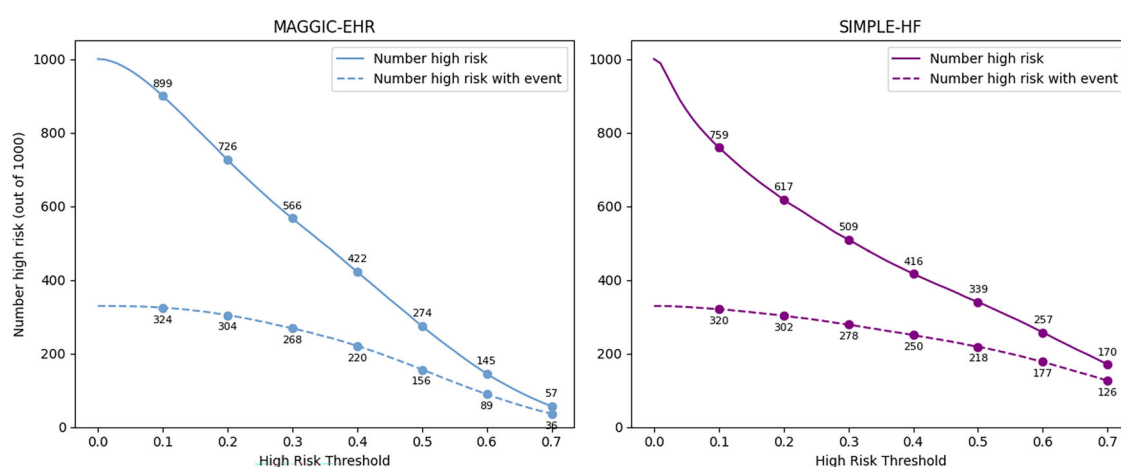
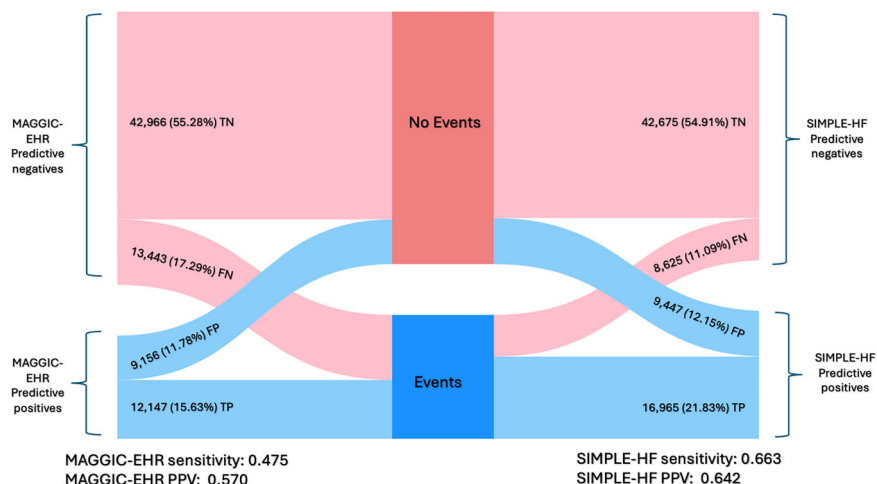


Fig. 4 | Clinical-impact curves for 12-month all-cause mortality prediction in the UK validation cohort. Comparison of MAGGIC-EHR (left) with the SIMPLE-HF model (right). For each possible decision threshold on the x-axis, the solid line

represents the number of patients per 1000 who would be classified as high risk, whereas the dashed line represents the subset of those high-risk patients who experienced the outcome (death). Numeric labels denote the corresponding absolute counts.

Discussion

In this study, we successfully developed and validated a parsimonious, 11-variable risk score that achieves high discrimination (C-index: 0.801, 95% CI [0.795–0.806]) for predicting all-cause mortality in patients with HF. Importantly, the model relies exclusively on variables readily available at the point-of-care, hence removing the need for specialised tests or longitudinal data streams.

The existing landscape of HF prognostication tools presents a dichotomy. Conventional risk scores, such as MAGGIC or the Seattle Heart Failure Model typically yield C-indices in the range of 0.61 to 0.75^{3,18}. These models have limited clinical utility in routine care settings due to their modest discriminatory properties and reliance on results of specialised testing procedures. This limitation is particularly significant since much of HF care in the UK and other countries occurs in primary care^{6,19}. Moreover, guidelines recommend re-assessment of risk across patients’ disease trajectory, resulting in ongoing reliance on specialised testing (such as subsequent echocardiography for LVEF measurements)⁴. Conversely, recent cutting-edge AI models leveraging dense, longitudinal EHR data can achieve C-indices exceeding 0.8¹³. However, their substantial data and computational demands restrict their widespread adoption and integration at the point-of-care^{15–17}.

The present work directly addresses this performance-practicality gap. Using a novel approach, we ‘distilled’ insights from a sophisticated Transformer-based AI model, initially trained on rich longitudinal EHR data, into a simpler MLP-based model. This process, guided by SHAP-based

feature selection, began with verifying that an MLP architecture using standard MAGGIC predictors could outperform the traditional Cox-based MAGGIC model. The subsequent augmentation with Transformer-identified comorbidity signals further improved the performance (Fig. S4). Lastly, feature distillation yielded the final 11-variable SIMPLE-HF model with gain of ~0.07 in C-index as compared to MAGGIC-EHR. This model not only surpassed the performance of an EHR-adapted MAGGIC benchmark but also substantially reduced the data collection burden by eliminating the need for variables like LVEF, which require specialised investigations. Moreover, the SIMPLE-HF includes variables related to comorbidities which is in-line with the current research that complex multi-morbidity, rather than HF itself, contributes to over 40% of mortality in HF patients^{9,20,21}. This highlights a significant gap in current risk assessment approaches, which SIMPLE-HF fulfils.

Beyond its immediate clinical relevance for the management of HF patients, this study also carries important methodological implications for the development of clinical prediction models. It showcases a pragmatic approach to harness the signal-finding capabilities of complex AI architectures, to inform the development of simpler and more implementable tools suitable for routine clinical use. Our findings suggest that the explainable AI techniques we used (SHAP values for feature selection in the present work and integrated gradients from our prior work for encounter-level attribution) can serve not only as interpretability tools for complex models, but also as powerful knowledge discovery and translation tools. Ultimately, this approach has the potential to bridge the gap between the

exploratory power of advanced AI and the pragmatic requirements of clinical decision support at the bedside.

The SIMPLE-HF feature set comprises 11 variables, all routinely collected during primary care encounters and broadly comparable in number and accessibility to widely used scores such as QRISK3²². None of these variables requires specialized testing or expensive investigations. While Transformer-based models incur substantial computational cost during training, this is a one-time cost; once distilled, the resulting MLP-based SIMPLE-HF model imposes minimal computational overhead and can be deployed on standard clinical hardware such as primary care or hospital laptops and tablets without dedicated AI infrastructure. A further advantage over MAGGIC is that SIMPLE-HF does not require NYHA classification, a subjective, clinician-assessed variable that is inconsistently recorded in routine EHR data across health systems internationally. Relying instead on objective, structured EHR variables (comorbidities, medications, demographics) supports the potential for broader external validation and adoption of SIMPLE-HF across diverse international settings, although such validation remains an important next step.

Several clinical and computational advantages are evident in the SIMPLE-HF score. Firstly, all 11 input variables are routinely collected during clinical encounters and require no specialised imaging, assays, or continuous longitudinal monitoring. In fact, they can also be compiled in the form of a checklist. Secondly, when using higher risk thresholds that are more relevant to clinical practice, the SIMPLE-HF model identifies more than twice as many patients who suffer mortality, compared to existing models. Third, our model is validated for prognostication across the entire HF trajectory, rather than being limited to the time of initial diagnosis. By employing random baseline selection, the model avoids bias toward any specific disease stage - whether early or advanced. This makes it more applicable to real-world clinical practice, where patients often present at varying points along the disease continuum. Collectively, these attributes suggest that the SIMPLE-HF score could be deployed even in healthcare settings with limited EHR interoperability, thereby supporting broader and more equitable access to robust risk stratification tools.

A further practical advantage of SIMPLE-HF is its avoidance of variables that are typically poorly recorded in routine EHRs. For example, NYHA classification was missing in approximately 95% of our cohort, which substantially limits the practical utility of any score relying on this variable. SIMPLE-HF instead draws on routinely captured comorbidities and current medications, which supports its applicability across diverse real-world settings, though further external validation in different healthcare systems remains necessary.

It is also worth noting that the high apparent prevalence of NYHA class II in our cohort (~46%) reflects the very high missingness of this variable in EHR data (~95% missing) and therefore should not be interpreted as representative of the typical hospitalised HF population. In practice, 12-month mortality prediction may have its greatest value not in long-term planning for stable patients, but in identifying patients with limited remaining therapeutic options who may benefit from earlier discussions regarding palliative or supportive care services, and in aligning treatment efforts to expected outcomes to avoid overtreatment in advanced disease.

This study has several notable strengths. These include the use of a large, nationally representative primary care dataset linked to hospitalisation and mortality records, a rigorous validation strategy employing a practice-level split, and the novel application of a distillation approach to develop a clinically practical tool. The improvement in discrimination over established benchmarks with reduced data requirements underscores the potential of our methodology.

However, certain limitations warrant consideration. Firstly, we acknowledge that this work demonstrates the validation of the pipeline within one cohort and one Transformer architecture; further external evaluation is required to determine generalisability across other settings, datasets, and architectures. Secondly, our adaptation of the MAGGIC score for EHR data (MAGGIC-EHR), while necessary and pragmatic to allow its implementation using routine clinical data, represents a modification of the

original score with several variables (e.g. New York Heart Association Classification (NYHA) or LVEF) missing to a significant extent in electronic health records, which should be considered when interpreting comparisons. Additionally, we substituted LVEF with a categorical variable representing HF subtype (reduced, preserved, or unknown LVEF). The 'unknown' category was substantial in our cohort, reflecting limited and inconsistent LVEF recording in routine primary care data. We acknowledge that this phenotyping approach is relatively recent and that further validation across cohorts is warranted. Lastly, our model's features were optimised for all-cause mortality; the same feature set was applied to the secondary outcomes without repeating the Transformer-guided feature discovery or SHAP-based distillation, which is reflected in the more modest decision-curve net benefit for those endpoints (Figs. S9–S12). Their analysis should therefore be viewed as exploratory, and optimised models for cardiovascular events or hospitalisation would require repeating the full pipeline for each endpoint.

To conclude, we translated the predictive signals identified by a complex Transformer model into an 11-variable risk score for mortality in HF that uses only routinely collected variables. With appropriate prospective validation, this approach may help support clinicians in risk stratification of HF patients. This could, potentially, include the identification of patients with limited 12-month survival who could benefit from earlier discussions regarding palliative or supportive care.

Methods

Data source and validation strategy

We used anonymised EHR data from the Clinical Practice Research Datalink (CPRD) Aurum and its linked datasets. Access to data from CPRD is subject to a full licence agreement containing detailed terms and conditions of use. Patient-level datasets can be extracted for researchers against specific study specifications, following protocol approval from the Independent Scientific Advisory Committee (ISAC). We used English data provided by the CPRD Aurum dataset with approval from the CPRD Independent Scientific Advisory Committee (protocol number: 20_095). The requirement for informed consent was waived by the ISAC because the CPRD data are pseudonymised. All research procedures were performed in accordance with the ethical principles of the Declaration of Helsinki.

CPRD provides detailed patients' records from primary care, including demographics, diagnoses, prescribed HF therapy, and health-related lifestyle variables, covering approximately 24% of the UK population, and broadly representative of age, sex, ethnicity, geographical spread, and socioeconomic deprivation^{8,23}. Linked data on hospital admissions and death certificates, were provided from Hospital Episode Statistics Admitted Patient Care (HES-APC) and Office for National Statistics (ONS), respectively, which are available for the subset of patients registered with English practices that have consented to data linkage and meet CPRD's linkage eligibility criteria^{8,23}.

We included data from 1442 contributing GP practices in England. To ensure robust validation, a subset of practices was randomly assigned to the derivation cohort for model development (80%—1153 practices), and the remaining to the validation cohort (20%—289 practices)²². This practice-level split mitigates data leakage by ensuring the model is tested on patients from practices it has never seen before. Additionally, this validation strategy provides a more rigorous assessment of model generalisability than splitting by individual patients, which can artificially inflate performance by testing the model on the same practice-specific data patterns it was trained on.

For MLP models, 5% of the derivation dataset was randomly selected to conduct end-of-epoch testing. This was done to avoid overfitting, which is a concern in AI parametric modelling. This testing, done at the end of every epoch (i.e., one full iteration of training in which the model has trained on all patients selected for training), ensured that the model is not overfitting on the derivation dataset (i.e., ultimately aiming to avoid data "memorisation"). After model training was estimated to have converged on the derivation dataset, the AI models were evaluated on the validation dataset. All patients in the derivation dataset were used for the fitting of statistical cox models (i.e., MAGGIC-EHR models).

Cohort selection

Included in the study were men and women registered with their general practice for at least one year during the study period (01/01/2010 to 31/12/2020) whose records were classified by the CPRD as acceptable for use in research and approved for HES and ONS linkage. This is based on recording and CPRD's internal consistency of key variables including date of birth, practice registration date and transfer out date. Among these, we identified an open cohort of individuals aged 18 years or older with incident HF. We identified HF diagnoses using previously published phenotyping algorithms for CPRD Aurum¹¹, which define HF using CPRD medical codes (medcodes, which map to SNOMED CT concepts) recorded in primary care and ICD-10 codes recorded in linked HES. Each patient's first record of HF was regarded as their incident diagnosis. We excluded individuals who had a HF diagnosis before the start of the study (01/01/2010), before their 18th birthday, or within the first 12 months of registration with their general practice.

Each patient was followed up starting at the latest of the following: January 1st 2010, their 18th birthday, 12 months after GP registration. Follow-up ended at the earliest of the following: date of outcome of interest (i.e., mortality), last collection date from practice, patient's last date in practice, or December 31st, 2020. For each HF patient included in the study, the index date (i.e., baseline), was randomly selected from the period between first HF diagnosis and end of study-period. This approach simultaneously ensures capture across the spectrum of calendar years and ages, and better reflects the reality of clinical practice, where patients are assessed at various stages of their care journey²⁴.

Outcome definitions

The primary outcome was all-cause mortality, at 12 and 36 months, determined from linked ONS mortality records.

To explore the broader utility of our approach, supplementary analyses were conducted to assess the performance in predicting two additional clinically relevant outcomes. These included (1) major adverse cardiovascular event, defined as a composite of fatal or non-fatal ischaemic heart disease, myocardial infarction, transient ischaemic attack, or stroke, and (2) any hospitalisation after incident HF. These outcomes were identified using both previously published and our own curated code dictionaries^{11,25}.

AI modelling and feature selection

All AI models were developed using a multi-layer perceptron (MLP) within a discrete-time survival framework using Ordinary Differential Equations Networks (SODEN)²⁶. Instead of maximum likelihood estimation (MLE) frameworks that utilise costly integral calculations for model training on censored data, the SODEN framework alternatively poses MLE as a differential-equation constrained optimisation objective²⁶. Furthermore, unlike previous proportional hazard frameworks that have not been shown to be theoretically robust for stochastic gradient descent (SGD) using mini-batching (i.e., random non-overlapping subsets of the derivation dataset), the SODEN framework alleviates many issues of established survival model frameworks that have trouble scaling to "big data". Specifically, with ordinary differential equations modelling the time-to-event distribution, the framework for MLE on censored data becomes more flexible (i.e., absent of strong structural assumptions of the shape of the survival/hazard distribution), more appropriately scalable (i.e., can be trained using SGD). The hyperparameters for the final MLP model are listed in Table S1.

Standard training protocols were implemented for all MLP models, including batch normalisation to stabilise learning, dropout for regularisation, and an early stopping criterion guided by the concordance index (C-index) on a held-out portion of the derivation dataset to prevent overfitting.

The different models underwent a development process involved iterative feature engineering and regularisation techniques to optimise performance and promote parsimony (Fig. S2). The steps included:

1. Baseline Model (MLP-MAGGIC): We began by developing a baseline model using the established MAGGIC predictor set. This initial model, named MLP-MAGGIC, served as a benchmark for subsequent

enhancements. These included sex, age, smoking status, diabetes, systolic blood pressure, creatinine, body mass index (BMI), New York Heart Association (NYHA) Functional Class, chronic obstructive pulmonary disease, any use of beta-blockers, any use of ACE inhibitor/Angiotensin receptor blockers, incident HF diagnosis ≥ 18 months prior to baseline, and HF subtype (i.e., preserved, reduced, or unknown according to previously published method)²⁷. The method assigns reduced or preserved LVEF on the basis of subtype-specific primary care medcodes and HES ICD-10 codes, with patients lacking any subtype-specific code classified as unknown²⁷.

2. Enriched Model (MLP-Explainability): To improve predictive power, we augmented the feature set. While our primary goal was parsimony in the number of features for clinical interpretability and ease of use, we were informed by findings from more complex models, specifically the Transformer-based TRisk model. This research highlighted the prognostic value of previously under-appreciated variables. This essentially included history of co-morbidities, namely, cardiac arrest, secondary malignant neoplasm of lung, hepatic failure, respiratory failure, pneumonitis due to food and vomit, secondary malignant neoplasm of bone/bone marrow, malignant neoplasm of lung, pneumonia, congestive heart failure, and acute renal failure. We identified the parent ICD code for each of these, i.e., the first three characters of the ICD-10 code (such as C78 for secondary malignant neoplasm of lung) and checked whether there was any code starting with that present in the history of a patient. The variables and the prevalence of them in our cohort is shown in Table S2. We included these into our initial MAGGIC set, subsequently with some feature engineering. This included combining the use of beta-blockers, and angiotensin-converting-enzyme (ACE) inhibitors/angiotensin receptor blockers (ARBs) (named "num_antihypertensives") and the cancers (identified by the explainability analysis on the transformer model – named "num_cancer"). This means if the patient does not have any cancer history, "num_cancer" would be 0, 1 if they have one of these cancers, and so on. Similarly for the anti-hypertensives as well.
3. We also incorporated the patient's year of birth as a proxy for birth cohort effects. This created an enriched model with 21 variables, named MLP-Explainability. (Table S2 for the exact explainability variables and their prevalence in the CPRD cohort).
4. Final Distilled Model (SIMPLE-HF): As a final step to enforce parsimony, we used SHapley Additive exPlanations (SHAP) to analyse the MLP-Explainability model and distil its 21 features down to only the most impactful predictors. This process resulted in our final model, SIMPLE-HF (Simplified Intelligent Mortality Prediction for Longitudinal EHRs in HF patients), which uses a refined set of just 11 variables: year of birth, occurrence of HF 18 months before baseline, age, cancer history, prescription of beta-blockers or ACE inhibitors/ARBs, body mass index (BMI), history of acute renal failure, pneumonia, cardiac arrest, respiratory failure, and chronic obstructive pulmonary disease (COPD). (Fig. S3 for results of SHAP analysis).

For all models, the predictors were captured from both primary care and HES records. Co-morbidities and medication use were extracted using previously published and curated phenotyping dictionaries for CPRD Aurum, based on primary care medcodes (SNOMED CT), HES ICD-10 codes, and product codes for prescriptions²⁸. For vital signs, such as systolic blood pressure, creatinine, and body mass index, average of records that were collected in the 36 months leading up to baseline were used as baseline variables for modelling. Last known smoking status in the 36 months leading up to baseline was used as baseline smoking status.

Benchmark modelling

For comparative purposes, the MAGGIC score was adapted for application to EHR data (termed MAGGIC-EHR), based on previously published methodology³. All predictors from the original MAGGIC score were extracted from EHR data, except for Left Ventricular Ejection Fraction

(LVEF), which is infrequently recorded in large-scale primary care EHR data. As a practical substitute for LVEF, we incorporated a categorical variable representing HF subtype (i.e., reduced, preserved, or unknown LVEF), derived using previously published phenotyping codes²⁷.

Missing values for systolic blood pressure, smoking status, body mass index, creatinine, and NYHA Functional Class were imputed for the MAGGIC-EHR model. Procedurally, for imputation of all-cause mortality prediction MAGGIC-EHR model, all variables for each model were included for imputation along with Nelson-Aalen estimate of baseline cumulative hazards for the outcome of mortality. Imputations were conducted on each derivation and validation datasets of our cohort separately using the “mice: Multivariate Imputation by Chained Equations” package in R²⁹. Following the extraction of derivation and validation datasets, the statistical models were fit on each of the five imputed derivation datasets with appropriate predictors (i.e., 5 MAGGIC-EHR models) for prediction of the primary outcome, all-cause mortality. The five Cox models were pooled using Rubin’s rules and evaluated on the 5 imputed validation datasets. The predictions were also appropriately pooled using extensions of the Rubin’s rules (i.e., complementary log-log transformations) and utilised for downstream analysis. The same procedures for model fitting and evaluation on validation datasets were repeated for the analyses of other outcomes: (1) fatal and non-fatal cardiovascular event prediction, and (2) rehospitalisation prediction.

It must be noted that the original MAGGIC model derivation study has solely investigated 12- and 36-month all-cause mortality³. To serve as the benchmark conventional model in this work, we used MAGGIC-EHR to not only predict 12- and 36-month all-cause mortality but also to demonstrate benchmark performance for the secondary outcomes investigated. Interestingly, while MAGGIC was not derived for secondary outcome investigations, validation studies have indeed been conducted for other outcomes such as CV-related mortality³⁰.

Performance analyses

We evaluated models’ performance on the UK validation dataset using multiple metrics. For discrimination, we assessed the concordance index (C-index) and area under the precision-recall curve (AUPRC). Model calibration was assessed graphically, by comparing the agreement between the observed and predicted risk using a smoothed calibration curve fit using restricted cubic splines with three knots, and quantitatively by calculating the integrated calibration index (ICI) metric³¹. In addition to this, we also conducted net benefit analyses at clinically relevant thresholds.

For all analyses, individual patient risk scores were derived by estimating survival probabilities over a 60-month follow-up period, focusing on the 12-, and 36-month timepoint. This is a characteristic of the SODEN model, which is able to estimate the survival probability at every month in the follow-up period²⁶.

Additionally, we conducted discrimination analyses in subgroups defined by sex, age, baseline systolic blood pressure, comorbidities and medication use at baseline.

Calibration and net benefit analyses for 12 and 36-month prediction were also conducted for all models to assess model performance at different follow-up durations.

Furthermore, we conducted an impact analysis to evaluate true positive and negative capture at the 50% decision threshold of the SIMPLE-HF and MAGGIC-EHR models on UK validation data in line with research recommendations by HF clinical guidelines⁶.

Reporting guidelines

For this research, we followed the TRIPOD-AI reporting guidelines to ensure transparent and rigorous reporting of our AI-based predictive modelling study³².

Data availability

CPRD data are available to researchers through licensing agreements following protocol approval by the Independent Scientific Advisory

Committee. Data are not publicly available due to licensing restrictions. Access and sharing policies are detailed on the [CPRD website](#).

Code availability

The TRisk model codebase is available at: <https://innovation.ox.ac.uk/technologies-available/technology-licensing/>. The data pre-processing is similar to that of the BEHRT modelling framework. The data pre-processing code, as well as the BEHRT model code is available here: <https://github.com/deepmedicine/BEHRT/tree/master>. Pre-trained model weights are available upon request of the corresponding authors. Interested researchers are welcome to contact the corresponding authors directly via the corresponding email address provided to obtain weights for research purposes.

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Author contributions

N.A. and S.R. conceptualised and designed the study. K.R. and S.R. provided supervision for the project. NA conducted all quantitative analyses. N.A. and S.R. wrote the first draft of the manuscript. N.A., S.R., N.C., B.O.P., M.W., Z.F., G.Z., K.R., and J.L. provided methodological insights on the manuscript. All authors were involved in revising the manuscript. N.A., S.R., and K.R. were responsible for the decision to submit the manuscript for publication. All authors have read and approved the manuscript.

Competing interests

K.R. is Editor-in-Chief of BMJ Heart. K.R. reports consulting fees from Medtronic CRDN and Lucem Health; honoraria or fees from BMJ Heart, PLoS Medicine, AstraZeneca Middle East and Africa Region, Medscape, Radcliffe Cardiology, and WebMD Medscape UK; and grants from the National Institute for Health and Care Research (NIHR), the Medical Research Council, the British Heart Foundation, the Novo Nordisk Foundation, Horizon Europe, and Roche. S.R. is a methodological advisor for BMJ Heart. S.R. reports consultancy fees from Lucem Health and grants from Oxford University Hospital Trust. N.C. reports grants from Wellcome Trust. The other authors declare no competing financial or non-financial interests.

Additional information

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